Ermiya Moghaddami

AKA: Ermia Moghadamy

Graphics/Systems Engineer
&

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SUMMARY

I'm **Ermia Moghadamy**. SUT Student, Software Engineer and ITVTO Instructor. I study how computers could work differently — graphics pipelines, system architecture, even analog computation. My work sits between engineering and research. I build things that don't just solve today's problems, but explore tomorrow's possibilities.
I'm looking for a research environment where ambition meets funding — and where a 19-year-old builder can become a world-class architect.

EDUCATION

Sharif University of Technology (SUT) | 2025-2029

Bachelor of Science in Applied Mathematics

- founder/leader of 2 Communities: SUT's MLC & Sharif Linux User Group

Organization for the Development of Exceptional Talents | 2019-2025

Diploma of Mathematics (Darolfonoon SAMPAD)

- GPA: 4.0
- Mentorship of students in coding festivals
- Participation in over 7 tech-related festivals of Ministry of Education
- Invited to Ministry of Education as young researcher figure (2023)

HONORS

Worldskills Medallion of Excellence (2025)

web technologies medallion of excellence along with participation as one of 5 candidates for WSC Asia Lyon (Taiwan) Competitions - affiliated to ***International Labour*** *Organization*

House of Research (2023)

ranked 2nd nationwide in Pazhoheshsara festival: Interactive web design.

Kharazmi Junior (2022)

ranked 1st nationwide in **Kharazmi** kharazmi mathematics major & hardware controller sub-major - *by Ministry of Education*

NAD Co. (2014-2015)

ranked 4th nationwide in **Robotics** competitions of **Early Elementary** School. at age 10 - *By* ***Sharif University of Technology***

PROJECTS

Zephrone (open-source)

A modern real-time 3D rendering engine built from scratch in Zig + OpenGL
(full info available on github)

HexHub (open-source)

customized simple implementation of MINIX alongside with DSL.
(full info available on github)

Mathbox (IoT)

game-box: capable of assisting students on arithmetic priority.
(artifact info available)

Hexiler (open-source)

A Low-Level LLVM Language Built for **HexHub**
(full info available on github)

CERTIFICATES

Instructor ID CARD

worldwide valid Instructor permit.
Iran's Technical & Vocational Training Organization Educator Permit License

Web Development

By Iran's Technical & Vocational Training Organization Skill License

Python Advanced Programming

By Iran's Technical & Vocational Training Organization Skill License

WorldSkills Certificate

WorldSkills Iran nation competitions participation certificate

NOTICE: "All TVTO credentials verifiable via serial number worldwide — *affiliated to* ***International Labour Organization***"

SKILLS

Soft Skills: Time Management, Team Work, Leadership, Community Organizing
Computer Graphics: Graphics Programming, OpenGL, wGPU, Vulkan, CUDA
System Programming: Operating Systems (MINIX), Compilers & Interpreters, DSL Design
Languages: Zig**, Rust*, Go, Python, TypeScript, SH, C, C++
Formal Proof & AFP: Haskell, Lean
DataScience & ML: Tensorflow, PyTorch, SciKit-Learn, NLP, Computer Vision
DevOps & IT: CI/CD and automation, SWE, Docker, Git, , Ansible, Linux Adminstration, Networks

EXPRIENCES

Iran TVTO Instructor (part-time)
TVT Organization: **"Nakhostin"** Institution Complex | 2022-**current**

- Educator: AI Fundamentals, AI Engineering, Programming, ML, Data Science, Software Engineer

Software Developer
"Fig" Startup Complex (including **FigAI**) | 2026-**current**

- Participation in Fig's "Development Team" at Fig's AI **Startup** from *Fig Complex*
- Founded by Mr. Amirparsa Baghdadi (worldskills Gold Medalist)

Software Engineer (project)
"Nabla" Academy | 2022-2024

- Designed, Developed & Maintained fully functional combined LMS, CMS and online exam platform completely customized due the preferences of Mr. Mahdi Mousavi (founder & Founder of **Nabla Academy)**.
- **stack:** Php, Laravel, Tailwind, Alpine, Livewire (TALL)

Software Developer (contract)
"Gravity" Institution Complex | 2025-2026

- Designed, Developed & Maintained fully functional combined LMS, online exam platform completely customized due the preferences of Mr. Muhamadreza Salehi (founder of**)**.
- **stack:** NuxtJs, TypeScript, Django, DRF, Tailwind

Software Engineer / Tech Support (contract / part-time)
"Hadaf" Clinic & Academy of Konkour / Psychology (Dr Marzban) | 2022-2024

- Designed, Developed & Maintained fully functional study tracking system with complex mentorship monitoring & live real-time contact of students with their mentor. in favor of Dr. Marzban & resulted in offical certificate of appreciation.
- **stack:** NextJS, Django, DRF, Bootstrap, Redux
- **challenges:** gently modifying business service architecture & System design very frequently

LANGUAGES

- **English** fluent / C1
- **Persian** native / bilingual
- **German** learning

NOTICE: "Available for interviews, technical assessments, or submission of supplementary materials upon request."

SOCIAL ACTIVITIES

Assistant Research Executive - Darolfonoon SAMPAD | 2021-2024

Appointed by school administration (research executive) to **mentor** student teams preparing for national and provincial competitions under the **Ministry of Education, including Kharazmi, Young Kharazmi, Abureyhan, and Research Center festivals**. Designed structured mentorship programs, guided project scoping and tech stack decision, reviewed and refined prototypes, and provided hands-on technical troubleshooting across multiple disciplines. Mentored students from multiple schools, including cross-institutional participants. Over three years, teams under mentorship earned 3 provincial and 5 nationalwide distinctions across 3+ fields. Entrusted with operational responsibility of the school's research office during project cycles.
***Associate Research Executive*:** Mr. Saeed Akbari Aghdam

Founding Lead — Sharif Machine Learning Community SUT | 2025–Present

Founded a structured peer-to-peer Machine Learning community after identifying a gap in disciplined self-directed learning among highly capable students. Recruited 60 active members and established biweekly technical sessions focused on conceptual depth and project-oriented learning. Designed a comprehensive GitHub-based learning roadmap and curated study resources. Led discussions while enforcing a rotating presentation system to ensure active participation and technical engagement from all members. Initiated early-stage collaborative research prototyping efforts; the community environment later influenced the development of my interest in engineering-oriented computer graphics.
I've also invited honor guests including WSC exceptional talents & exprienced engineers

Founding Lead - Sharif LUG (Linux User Group) | 2025-Present

Identified a critical gap in Linux literacy within the CS/Math Campus, where no active LUG existed despite strong technical culture. Took initiative to personally onboard 5-7 new Linux users in one week through one-on-one mentoring and conversations. **founded/led** SharifLUG with the vision of using Linux as a gateway for continuous learning—each event structured around discovering one new technical concept (team work, project management, version control systems, containerization, automation, system hardening, etc.), not just OS advocacy.
Impact: Revived local open-source culture at one of Iran's top engineering school MathSci Campus.

RESEARCH & PUBLICATIONS

Research Interests:

- Computer Graphics Engineering
- System Engineering
- Logic & Computation
- Alternative Computation

Letter of Recommendation:

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Currently Researching:

- **PROPOSAL:** Evaluating Foundation Models in Low-Level Graphics Code Generation: *Empirical Challenges and a Robust Abstraction Framework*
- **PROPOSAL:** ORB Graphics computations: Implementing an optimized GPU-based computation alternative algorithm

NOTICE: "Proposal Preprint available upon request."